

Saumya Dwivedi

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Full-stack engineer building scalable applications and automated workflows. Proficient in Node.js, Next.js, Python, React, and PostgreSQL, with a proven track record of deploying production-grade architecture and integrating AI/LLMs.

EDUCATION

VIT Bhopal University

B.Tech in Computer Science (9.07 CGPA)

Bhopal, MP

Sep 2023 – May 2027

The Jain International School

Intermediate, CBSE (86.4%)

Mar 2023

Ryan International School

Matriculation, CBSE (96.66%)

Mar 2021

EXPERIENCE

FOSSEE Summer Fellow

May 2026 – Present

IIT Bombay

- Selected through a competitive national-level screening process from across India.
- Architected Kubernetes deployment infrastructure enabling the Django, Celery, PostgreSQL, and Redis services platform to scale simulation workloads reliably across multiple tenants, delivered through Agile sprints.
- Building a dynamic pod orchestration system with a custom pod manager to provision ephemeral simulation pods from templated specs, reducing manual setup overhead for new simulation workloads.
- Implementing CPU/memory resource limits across pods to ensure workload isolation in a multi-tenant environment, preventing any single simulation job from degrading platform-wide performance.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, Java, SQL, HTML5, CSS3

Frontend: React.js, Next.js, HTML5, CSS3, React Flow

Backend: Node.js, FastAPI, Django, Flask, REST APIs, JWT Authentication

Databases: PostgreSQL, MongoDB, MySQL, Redis

DevOps & Tools: Git, CI/CD, GitHub Actions, Docker, Kubernetes

AI & ML: Gemini API, LLM Integration, PyTorch, TensorFlow, scikit-learn, SHAP

PROJECTS

Archon - AI Codebase Intelligence Platform | Next.js, Node.js, MongoDB, React Flow, Gemini API

[GitHub](#) | [Live Demo](#)

June 2026

- Large codebases hide structural risks that generic AI cannot parse; built Archon to instantly convert any GitHub URL into a fully parsed, visualized, and AI-explainable dependency map.
- Engineered a custom static-analysis pipeline (AST parsing, DFS cycle detection, complexity scoring) rendered as an interactive React Flow graph with dynamic layouts and click-to-focus isolation for 300+ file repositories.
- Integrated Gemini for graph-grounded codebase chat and fix suggestions, backed by an async pipeline (Socket.IO, JWT/OAuth) and deployed end-to-end on Render with GitHub Actions (CI/CD) and Uptime Robot.

ChurnGuard – Customer Churn Prediction System | FastAPI, Streamlit, MongoDB, SHAP, scikit-learn

[GitHub](#) | [Live Demo](#)

Mar 2026

- SaaS businesses typically catch churn only after revenue is unrecoverable; engineered Churn Guard to flag at-risk customers early and automate proactive intervention workflows for business teams.
- A raw probability score isn't actionable; integrated SHAP explainability and optimized model thresholds (AUC 0.835) to surface clear, per-customer churn drivers through a scalable Streamlit UI.
- A churn score alone tells a business team nothing actionable, layered in SHAP explainability to surface the top drivers behind each prediction, turning a bare number into a clear, per-customer reason teams can act on.
- Designed a production-grade FastAPI backend for bulk scoring (1000+ customers per API call), featuring robust REST APIs, strict validation, and MySQL integration for seamless, high-performance deployment.

CERTIFICATIONS

ServiceNow Virtual Internship Program: ServiceNow University | [View Certificate](#)

Computer Networking: The Bits and Bytes of Computer Networking – Google | [View Certificate](#)

DP-900: Microsoft Azure Data Fundamentals | [View Certificate](#)

NPTEL: Introduction to Machine Learning | [View Certificate](#)